

Little Red Dots: Super-Eddington Winds, Not Overmassive Black Holes

arXiv:2606.30711 — Little Red Dots as intermediate-mass, super-Eddington engines.

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Type IIIn supernovae and eta Carinae's Great Eruption offer a stellar template for JWST's enigmatic Little Red Dots.

Background

JWST's Little Red Dots show a unique combination of features: blackbody-like continua, broad Balmer lines, and stellar absorption, unmatched by any other galaxy or AGN class.

Their nature is debated; standard AGN models struggle to explain the peculiar spectra and inferred overmassive black holes.

What this paper is selling

The authors argue LRDs resemble scaled-up Type IIIn supernovae and eta Carinae's Great Eruption, where dense winds create pseudo-photospheres. This wind physics may mimic a virial broad-line region, biasing black hole mass estimates.

Key results to remember

- Dense wind pseudo-photospheres at ~5000 K explain LRD continua and line profiles.
- Escape velocity argument yields engine masses below 10^5 solar masses.
- Super-Eddington intermediate-mass engines may seed massive black holes.

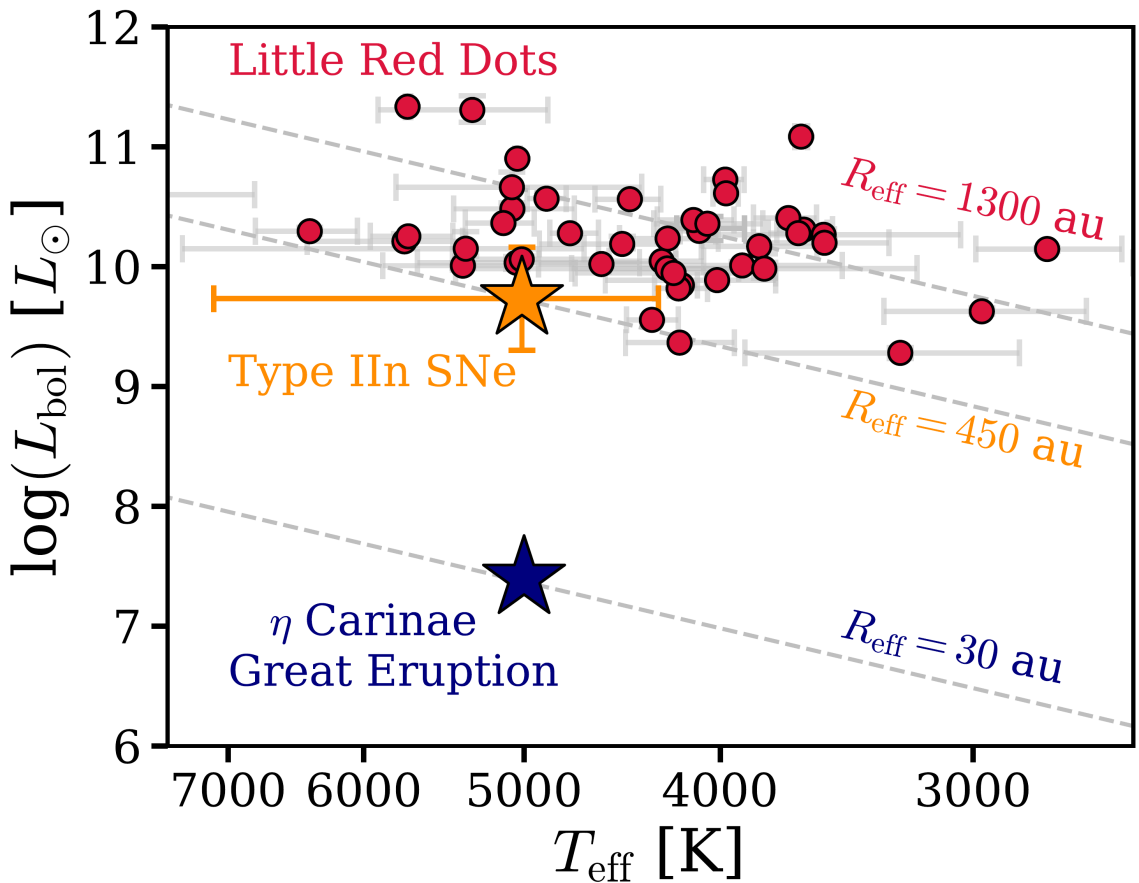


Figure 4 | LRDs, Type IIIn SNe, and the GE span 4 dex in bolometric luminosity (L_{bol}) but cluster in effective temperature (T_{eff}).

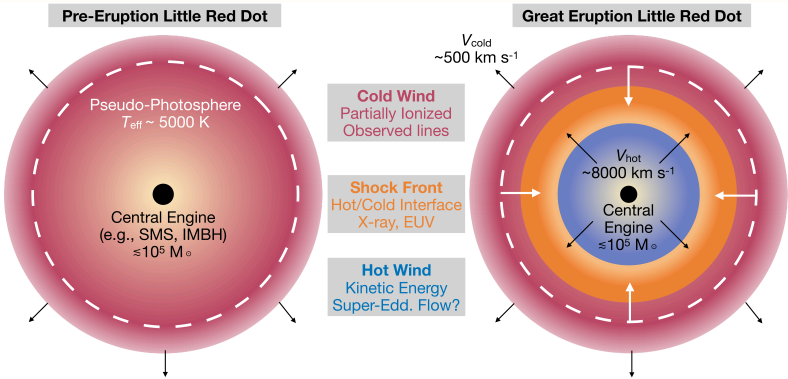


Figure 8 | Schematic of the proposed LRD scenario inspired by Type IIIn SNe and the GE.